ISSUES IN MODERN PRODUCTION SYSTEMS

Dr. Maria Papadakis | papadamc@jmu.edu | King Hall 305 | Mobile 540.746.7064
office hours | Mon & Wed 3:00-5:00 pm | and by appointment

I respond to email from 9 am to 6 pm. If it is time-sensitive, text me between 10 am – 9 pm, any day including weekends.

Dr. Hao Zhang | zhang9hx@jmu.edu | EnGeo 2110 |
office hours | Tues 1:00-2:00 pm and 3:40-4:40 pm | and by appointment

Required Materials

- Earbuds/headphones for lab
- Safety glasses for lab
- Free Fusion360 software (please wait for instructions)

Course Themes

1. The Nature of Production
2. Manufacturing Efficiency and Costs

Lab King Hall 146

Section 1: Dr. Papadakis
Th 2:20 - 3:35 pm

Section 2: Dr. Zhang
Th 3:55 - 5:10 pm

Section 3: Dr. Zhang
Mo 3:00-4:15 pm

Mark Showalter, lab manager
- showalmh@jmu.edu

TAs: See Canvas. Varies by section.

Overview

Welcome to ISAT 211! This is a course that we hope you will enjoy. Our focus is on the technological and economic management of manufacturing systems. Or, very simply, "how we make stuff." ISAT 211 is a required course in the ISAT curriculum.

The "big idea" of ISAT 211 is that *everything flows*. Manufacturing engineering, operations, and management are all based on our ability to analyze flows of materials and transformational operations at many scales – machine, production line, factory, and supply chain. Because the world of manufacturing is a perfect intersection of technology and economics, we will explore how decision-making in these two worlds interact with one another.

Broadly, the purpose of ISAT 211 is to give you foundational knowledge about how products are engineered, physically manufactured, and managed in a production process. This purpose derives from the broader ISAT program goals, and our class will reinforce your abilities in system thinking, sequential logic, mathematics, sensitivity to social context, and principles of technology. Detailed learning objectives are provided in the weekly *Topic Study Guides*.

Assignments and Grade Weights

TESTS: 50%

- Test 1, 25%
  Week of March 4 in your assigned lab period
- Test 2, 25%
  Monday, May 6
  1-3 pm in our classroom

EXERCISES AND LABS: 25%

Labs are activities that are usually (but not always) completed during lab time. You will also have several graded exercises. Exercises are applications of skills you are learning.

Makery Project: 25%

A project in which you design and make a product prototype. We'll give you a couple basic things to choose from, which you can adapt. You will work on it throughout the semester, but the physical prototyping will happen after Spring Break.

HOW TO BE SUCCESSFUL

Show Up.

Do the Work.

Get A Study Buddy.

Request/use your disability accommodations.

Key Policies

ATTENDANCE & EXTENDED ABSENCE

Attendance it is not required for lecture. It ***is*** required for lab to get credit for a lab assignment, *unless* we give you an accommodation *in advance* (or an emergency prevents you from contacting us ahead of time).

If you have to miss class for a compelling reason, we will work out accommodations on assignments you might miss. However, if you are absent more than 2 lectures or 1 lab/1 lecture consecutively, you must notify the Dean of Students about your extended absence (link on Canvas). After we have been contacted by the Dean, then we can work out a plan with you for late work.

LATE & MAKE-UP WORK

For other than an extended absence, late work for any assignment is allowed *only if you have worked out an accommodation with us in advance* or an emergency prevents you from doing so. Life happens— crappy stuff (deaths in the family, going to jail, bad romance) and good stuff (job interviews, celebrations, 21st birthdays). Real life gets real accommodation in our world, but please don't abuse this professional courtesy.

In advance can literally mean minutes before an assignment is due. You can expect one accommodation from us with no questions asked. You don't have to give a reason, so you don't need to worry if we will or won't give you a break the first time you ask. You only have to say "I really need more time on XYZ right now." Our accommodation is to allow you to turn it in late (at a mutually agreed upon time) without penalty. We guarantee only one such accommodation. *If you are having difficulty keeping up with life and school, we will need to chat to best sort it out*. Who should you contact...?

- ➡ Dr. P for tests and exercises
- ➡ Your assigned lab professor for labs or the Makery Project

GENERATIVE AI (gAI)

ChatGPT and generative AI tools are very helpful. However, you are not allowed to use them on any assignments in this class. The main reason is that the assignments aren't really gAI friendly.

BAD WEATHER CLOSINGS

We won't have class or lab if JMU closes, period. Whether we meet on the official make-up day depends on where we are in the course and if we can stay on track without it.

HONOR CODE

The JMU honor is in full effect for our class. No help is allowed on tests. Collaboration on other work is explained for each assignment.

COURSE GRADING SCALE: NOTE THE SMALL "D" RANGE

A: 91-100 A–: 90 B+: 88-89 B: 81-87 B–: 80
C+: 78-79 C: 71-77 C–: 70 D: 66-69 F: below 66

You need a C+ in ISAT 211 to be eligible for the Global Supply Chain Minor.